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Instruction

What the question is

My Notes:

TM Variants

Sketch a state diagram of a TM to recognize each of the languages described in: 3.8b (Assuming a leading blank)

UHH WE ALREADY DID THIS IN THE PREVIOUS ASSIGNMENT. AND IT HASN'T BEEN GRADED YET SO…??? I'm gonna copy and paste that because what????

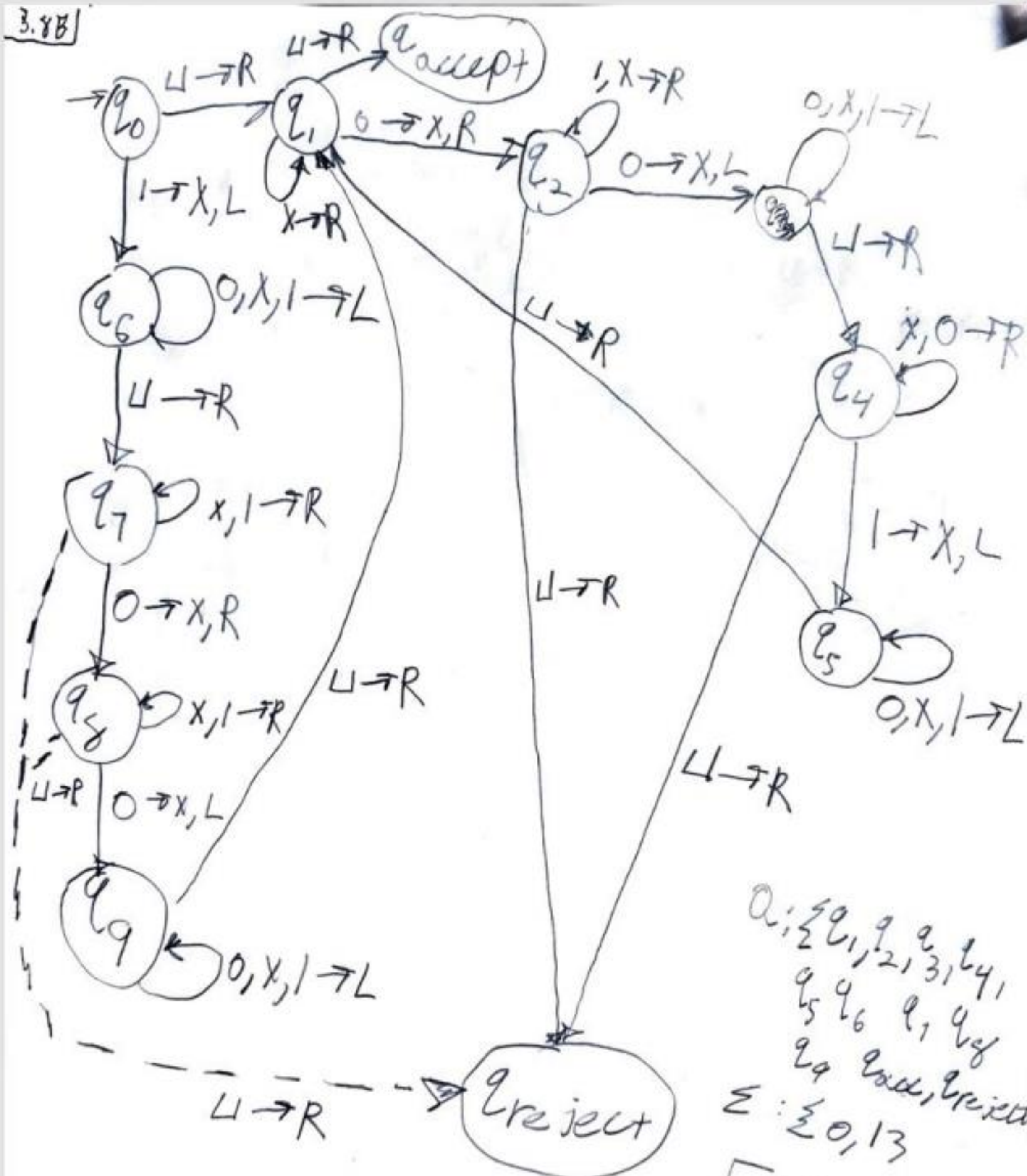
3.8 Give implementation-level descriptions of Turing machines that decide the following languages over the alphabet $\{0,1\}$.

[redacted]

b. $\{w \mid w$ contains twice as many 0s as 1s$\}$

[redacted]

3.8B



$Q: \{q_1, q_2, q_3, q_4, q_5\ q_6\ q_7\ q_8\ q_9\ q_{acc}, q_{reject}\}$

$\Sigma: \{0, 1\}$

$\Gamma: \{0, 1, X, \sqcup\}$

$q_0: q_0$

$q_{accept}: q_{accepts}$

$q_{reject}: q_{reject}$

$\delta: Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$

credit to Office hours since I didn't know where to start

TM Variant: 3.11

3.11 A ***Turing machine with doubly infinite tape*** is similar to an ordinary Turing machine, but its tape is infinite to the left as well as to the right. The tape is initially filled with blanks except for the portion that contains the input. Computation is defined as usual except that the head never encounters an end to the tape as it moves leftward. Show that this type of Turing machine recognizes the class of Turing-recognizable languages.

DID THIS PREVIOUSLY

3.11 To prove the double infinite tape T.M D = the ... T.M M. To show equivalence between D & M we need to show 2 things

- Any Lang L that can be recognized by D can be recognized by M
- Any Lang L that can be recognized by M can be recognized by D as well.

1) Sim M by D
 1) Mark left hand end of D
 2) Prevent D from moving it's head to the left of mark
 D sim M.

2) Sim D → M:
To sim the double ∞ tape Turing machine D by an ordinary TM M, sim it w/ a 2 tape TM.
As the 2 tape T.M was already equiv is pow to an ordinary T.M
Sim of 2X ∞ tape T.M D w/ 2 tape TM M,
1st tape of M, is written w/ input str & the 2nd tape = blank
- Cut the tape of 2X ∞ tape T.M D into parts, at the starting cell of input str.
 1 pt has input str & all blank space = right
 2nd pt containing left of the input str appears on the 2nd tape in reverse order.
 M sim D.
Thus 1 & 2
 D is equiv to M. T.M w/ 2X ∞ tape is equiv to an ordinary T.M

TM Closure: 3.15c, 3.15d, 3.15e

is Turing-recognizable.

3.15 Show that the collection of decidable languages is closed under the operation of

[redacted]

d. complementation.

e. intersection.

c. star.

AGAIN PREVIOUSLY DID ON OTHER ASSIGNMENT

3.15(c) Let L be a Turing decidable Lang. & M be the T.M that decides L.

There's a T.M M' such that, it decides of L i.e, $L(M') = L^*$

Desc of M' is:

M' = on input w:

1. Split w into n pts such that $w = w_1 w_2 \dots w_n$ in different ways.
2. Run M on w_i for $i = 1, 2, \dots, n$.
   If M accepts each of these str $w_i \rightarrow$ accept
3. All cuts have been tried w/out success then reject.

When w is cut into differ substrings such that every str. is accepted by M, then w belongs to the star of L & thus M' accepts w after finite # of steps, else w will $\rightarrow$ reject, b/c there are finitely many possible cuts of w, M' will halt after finitely many steps.

Therefore, $L(M') = L^*$. The decidable Lang. are closed under star.

3.15d For a turing decidable Lang. L, T.M decides Lang M then the complement is M' on input w.
The desc of M' is as follows
M' = on input w:
1. Accepts if M rejects
2. Else accept
Since M' Does the opposite of whatever M does, it decides the complement of L.
Therefore, decidable Lang. are closed under complementation.

3.15e Let L_1 & L_2 be 2 turing decidable Lang
M_1 & M_2 be the T.M that decides L_1 & L_2 respectively
There is a T.M M' such that, it decides intersection
Of L_1 & L_2 ie $L(M') = L_1 \cap L_2$
Desc of M'

M' = on input w;

1. Run M_1 on w. If M_1 rejects then reject.
2. Else run M_2 on w. If M_2 rejects then reject
3. Else accept.

M' Accepts w if both M_1 & M_2 accepts, if, if either
Of the rejects then M' rejects w.
Thus, $L(M') = L_1 \cap L_2$. The decidable lang
are closed under intersect.